Problem 5

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Problem 1 *A table of values for f and f' is shown below. Suppose that f is a one-to-one function and f^{-1} is its inverse.*

x	$f(x)$	$f'(x)$
1	*3*	*4*
3	*4*	*5*
4	*6*	*3*

(a) *Evaluate $f^{-1}(f(x))$ at $x = 3$.*

Explanation. $f^{-1}(f(3)) = f^{-1}(4) = 3$

(b) *Evaluate $\frac{d}{dx}f(f(x))$ at $x = 3$.*

Explanation.

$$\frac{d}{dx}f(f(x)) = f'(f(x)) \cdot f'(x) \implies \left[\frac{d}{dx}f(f(x))\right]_{x=3} = f'(f(3)) \cdot f'(3)$$
$$= f'(4) \cdot 5 = 3 \cdot 5 = 15$$

(c) *Evaluate $\frac{d}{dx}\ln((f(x))$ at $x = 3$. [2]*

Explanation.

$$\frac{d}{dx}\ln((f(x)) = \frac{f'(x)}{f(x)}$$
$$\implies \left[\frac{d}{dx}\ln(f(x))\right]_{x=3} = \frac{f'(3)}{f(3)} = \frac{5}{4}$$

(d) *Evaluate $f^{-1}(x)$ at $x = 3$. [2]*

Explanation. $f^{-1}(3) = 1 \iff 3 = f(1)$

(e) *Evaluate $\frac{d}{dx}f^{-1}(x)$ at $x = 3$. [2]*

Explanation.

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$$
$$\implies \left[\frac{d}{dx}f^{-1}(x)\right]_{x=3} = \frac{1}{f'(f^{-1}(3))}$$
$$= \frac{1}{f'(1)} = \frac{1}{4}$$

(f) *Evaluate* $\lim_{x \to 4} \frac{f(x) - f(4)}{x - 4}$ *[2]*

Explanation. $\lim_{x \to 4} \frac{f(x) - f(4)}{x - 4} = f'(4) = 3$